

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTEEPLE504B - POWER QUALITY ISSUES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which of the following is a common source of Voltage Sags?
a) Lightning strikes on transmission lines. b) Starting of large induction motors.
c) Line-to-ground faults. d) All of the above
 - 2 The primary difference between a 'Swell' and an 'Overvoltage' is:
a) The magnitude of the voltage. b) The duration of the event.
c) The cause of the event. d) There is no difference.
 - 3 Which device is commonly used to clamp transient overvoltages?
a) Isolation Transformer b) Metal Oxide Varistor (MOV)
c) Circuit Breaker d) Fuse
 - 4 'Ferro-resonance' is a phenomenon characterized by:
a) Interaction between system capacitance and non-linear inductance (transformer core).
b) Interaction between resistance and capacitance.
c) A short circuit fault.
d) Lightning strikes.
 - 5 A Surge Arrester should be placed:
a) As far as possible from the equipment to be protected.
b) As close as possible to the equipment to be protected.
c) Only at the generation station.
d) In series with the load.
 - 6 IEEE Standard 519 primarily provides guidelines for:
a) Voltage sags. b) Lightning protection. c) Harmonic control in electrical power systems. d) Grounding.
 - 7 Which index is most commonly used to measure harmonic distortion in voltage?
a) TDD (Total Demand Distortion) b) THD (Total Harmonic Distortion) c) RMS d) Crest Factor

- 8 'Flicker' is defined as: a) A permanent blackout.
b) A subjective impression of fluctuating luminance caused by voltage fluctuation.
c) A harmonic distortion. d) A frequency deviation.
- 9 Pst and Plt are indices associated with:
a) Harmonic Distortion. b) Transient measurement. c)
Flicker severity measurement. d) Frequency stability.
- 10 Which is NOT a standard voltage quality parameter to monitor?
a) Frequency b) Unbalance c) Harmonics d)
Wind speed
- 11 Smart Power Quality Monitors (Intelligent Systems) can:
a) Automatically classify and analyze events b) Repair the fault
automatically
c) Generate electricity d) Replace circuit
breakers
- 12 A sudden, non-power frequency change in voltage or current is known as:
a) Harmonic distortion b) Voltage imbalance c) Transient
d) Voltage fluctuation
- Q2. Solve the following.
- A) Define power quality. Explain why power quality is important? 6
- B) With a waveform sketch, explain the terms(a)Voltage sag(b)Voltage interruption(c)Voltage swells 6
- Q3. Solve the following.
- A) Explain the sources of sags in power system 6
- B) Explain sources of transient overvoltage. Explain any two devices for over voltage protection. 6
- Q4. Solve Any Two of the following.
- A) Explain any four devices to maintain voltage regulation 6
- B) Write short notes on THD and TDD. 6
- C) Explain different flicker mitigation techniques. 6
- Q5. Solve Any Two of the following.
- A) Briefly discuss the common objectives of power quality monitoring. 6
- B) Define flicker and explain sources of flicker. 6
- C) Explain the long duration voltage variations. 6
- Q6. Solve Any Two of the following.
- A) State and explain selection criteria for power Quality Monitoring 6
- B) List various types of power quality measuring equipment and explain any three type power quality measuring system. 6
- C) Explain the steps involved in power quality monitoring. What are the information from monitoring site surveys? 6

*** End ***